

Recently, an organisation was in the news for offering notebooks to a majority of their employees. While to most people, this may seem like another example of corporate indulgence, there is in fact, calculated economics behind the move. A normal office PC costs anywhere around Rs 35,000, whereas an entry-level laptop costs at least twice as much. By giving the employee, the employer is assured of work being done anywhere—be it in transit or at home—beyond the normal working hours. The place simply doesn't matter. As long as the employee has the device with him or her, he or she will work. On the other hand, deployment costs in terms of networking are reduced, and less desk space is used. Recently, many Indian business schools made it mandatory for their students to have a laptop before they enrol themselves for the course.

The world is going mobile in the true sense. Despite the

availability of laptops, the acceptance rate in India is far less as compared to the West or, for that matter, our neighbouring Eastern countries. The recent move by many established international vendors such as IBM, Acer and HP to provide entry-level laptops below Rs 50,000 will further increase the growth in this segment.

Coming back to our comparison tests, we did our best to cover as many products from different vendors. Reputed international brands such as IBM, HP Compaq, Toshiba, Fujitsu and Acer took part in the comparison. We also included many Indian-based manufactures such as ACi, Micro-D, BP—with its distinct looking models—and the newly launched PnP. We managed to get one laptop from Wipro's peripheral division, and of course, Zenith. Dell only managed to send in one laptop, the Inspiron 1100. We did not include Sony's VAIO range since it did not have any authorised Indian vendor.



How We Tested

Laptops were assessed on four main parameters—features, performance, usability and package content. Their weightages varied from one category to another. The features weightage remained constant at 35 per cent across all categories, but as each category has different requirements, each individual feature had varied weightage across categories.

We used Windows XP as the standard operating system. Each laptop was formatted, and a fresh copy of Windows XP professional was installed. After the installation, all the latest drivers were loaded, without any manufacturer provided tools or customised software. All Tablet PCs were tested on Windows XP Tablet edition. For the temperature test, we used a thermistor-based thermometer to note the temperatures. The ambient temperature was kept constant at 26 degrees Celsius, before commencing the test. The laptops were weighed on an electronic scale for accurate weight measurement.

Features

Features are of key importance to laptops, as they act as the sole differentiators between models. Physical features were not considered under this parameter, but were assessed under usability. We noted various features such as the type of RAM, hard disk capacity and speed, type of optical drive provided, types of ports, connectivity options, etc. Each of these was then scored according to the importance they played in the category the laptop was placed in. Processor speeds and amount of memory were not given any weightage as the extra speeds should be reflected in the performance tests.

Usability

A laptop's usability is evaluated on the basis of how simple or difficult it is to handle in day to day life—whether it has shortcut keys, its weight, etc. We also subjectively weighed the build quality—three different people were asked to rate the build quality on a scale of five, and the average was calculated. A similar exercise was used to rate the overall ergonomics of the laptop.

To assess the speaker quality, we played music files that encompassed the complete audio frequency range. Based on the perceived output, we rated them on a scale of one to five. Since laptops are mobile work companions, it is essential to log the maximum heat dissipated by them. We ran a movie for 20 minutes to

simulate the high activity scenario, and noted the temperature on the under belly of the laptop.

Package contents

We noted whether the vendor provided the operating system with the laptop; whether they bundle a recovery CD, users manual and driver CDs. We also noted the extra software provided, and the accessories—power adapter, carry case, etc.—bundled along.

Performance

To gauge the performance of each laptop, we ran a battery of tests to evaluate each individual sub-system. The following benchmarking suites were used to test the laptops:

Si Soft Sandra 2003 professional: This benchmarking suite was used to evaluate CPU, memory and hard disk performances. The CPU benchmark gives us the arithmetic and multimedia scores for a particular CPU. The hard disk benchmark yields sequential read, sequential write, random read and random write speeds of a hard disk, as well as an estimated access time.

Ziff Davis Business Winstone 2002: ZD bench's Business Winstone offers an excellent way of evaluating the complete system performance, and returns a unified score that is an aggregate of the individual tests—the higher the score, the better the system performance. This benchmark suits the purpose because it tests a system using applications that are used on a daily basis; it simulates real world usage to arrive at the final score, which reflects how the system will fare under real life loads.

Video Encoding: We encoded a 50 MB video file using Virtual Dub to convert an MPEG file to DIVX. The codec used for compression was DIVX 4. The time taken to compress the file was noted—the faster the CPU and memory subsystem, the faster the encoding of the file.

Quake III: Arena and 3D Mark 2001 SE: As most of these systems come with low-end graphics adapters, using heavy graphic benchmarking tools doesn't make much sense. We relied on *Quake III: Arena* to test the graphics performance, and logged the frame rates at different settings. We also used 3Dmark 2001 SE to further assess the graphics capabilities.

MobileMark 2002: We used MobileMark to evaluate the battery life, which simulates real world usage. It drains the battery from 100 per cent until the system cannot handle the load anymore,